

GEOLOGIC AND ENVIRONMENTAL HAZARD

Mines

Old mines can leave hazards long after production ends: open shafts, unstable ground, tailings, acid drainage, metals, and unrecorded workings.



1 Understand the factor

What it is - and why it matters

Mine-related features include active and abandoned pits, underground workings, adits, shafts, waste-rock piles, tailings, mills, smelters, haul roads, and drainage tunnels. Hazards vary by commodity and geology. Physical dangers include falls, collapse, subsidence, open water, explosives, and unstable waste piles. Environmental concerns can include acidic drainage, arsenic, lead, mercury, cadmium, asbestos, uranium, or other materials in soil, dust, sediment, groundwater, and surface water. Many historic features were poorly mapped or predate modern reclamation standards.

How risk can reach a household

- Underground voids or shallow workings can collapse, creating subsidence, sinkholes, cracks, or loss of support.
- Tailings and waste rock can erode or generate contaminated dust and runoff, particularly during storms or disturbance.
- Mine drainage can mobilize metals and acidity into streams, wells, wetlands, or irrigation water.
- Open shafts, highwalls, pits, unstable structures, and water-filled workings present immediate injury and drowning hazards.

What people often miss

- A reclaimed-looking landscape may still contain buried openings, contaminated fill, or long-term drainage systems.
- Mine maps can be incomplete, inaccurate, or held in separate federal, state, county, and historical archives.
- Metal concentrations are naturally variable; sampling must distinguish background geology from mine-related contamination.
- Wildfire, flooding, off-road vehicles, construction, or illegal dumping can damage covers and expose mine waste.

? Five checks before buying

1

Search federal and state abandoned-mine inventories, geological maps, mineral records, historical aerials, and title documents.

2

Have a geotechnical professional assess suspected underground workings, subsidence, pits, fills, highwalls, and foundation implications.

3

Use qualified environmental sampling for soil, dust, water, tailings, and vapor when a credible pathway exists.

4

Determine whether reclamation obligations, access, monitoring, water-treatment, or land-use restrictions attach to the parcel.

5

Ask the insurer and lender about mine subsidence, earth movement, pollution, attractive-nuisance liability, and private roads.

IMPORTANT DISTINCTION Map proximity is a screening signal - not proof of exposure, damage, or insurability. Confirm the exact feature, current condition, pathway, and property-specific controls.

HEALTH, PROPERTY AND COVERAGE

How this factor can affect a household

FACTOR 12

H Health impacts

- Falls, collapse, drowning, unstable structures, and unexploded materials can cause severe injury or death.
- Acidic or metal-rich water may irritate skin and can make a private well unsuitable without treatment or replacement.
- Chronic ingestion or inhalation of contaminated soil, dust, or water can affect the nervous system, kidneys, lungs, and development, depending on the contaminant.
- Visible hazards, warning signs, and uncertainty about children or pets entering mine areas can create persistent stress.

P Property impacts

- Subsidence can crack foundations, roads, utilities, wells, septic systems, retaining walls, and building pads.
- Water treatment, shaft closure, fencing, stabilization, and reclamation can be technically complex and expensive.
- Tailings and contaminated fill can complicate grading, gardening, drainage, excavation, disposal, and construction permits.
- Known mine hazards can affect usable acreage, disclosure, appraisal, financing, marketability, and emergency access.

I Insurance implications

- Homeowners insurance usually excludes earth movement, settling, land, pollution, and pre-existing mine conditions.
- Sudden collapse damage may still be disputed based on policy wording and whether the cause is excluded earth movement.
- Some states or areas offer mine-subsidence insurance, but eligibility, covered causes, limits, and deductibles vary.
- Liability coverage may be jeopardized if known shafts, pits, or unstable areas are not secured and someone is injured.

Possible long-term health implications of buying nearby

- Long-term health risk is highest when mine-derived contaminants enter drinking water, household dust, garden soil, or play areas.
- Private wells require contaminant-specific testing because standard basic panels may not include mine-related metals or radionuclides.
- A mapped mine nearby may create no residential exposure if it is stable and isolated; an unmarked tailings pile on the lot can be far more important.
- Ongoing drainage treatment or restricted land use can influence household routine and costs for decades.

Evidence lens

Mine collapse and contaminant mechanisms are well established. Site-specific engineering, historical research, and laboratory testing are needed because old mine records and material distribution are often uncertain.

Plan more carefully for

Children

Older adults

Pregnancy

Heart/lung conditions

Mobility or power needs

Insurance reality: Availability, eligibility, exclusions, deductibles, and limits vary by state, carrier, property, and cause of loss. Get an address-specific written quote before the purchase contingency expires.

PRACTICAL RISK REDUCTION

Precautions and a real-life example

FACTOR 12

V Verify the actual risk

- Search multiple mine and geology databases and walk the parcel with professionals familiar with local mining history.
- Locate shafts, adits, tailings, drainage, old foundations, subsidence, warning signs, and any recorded reclamation work.

P Protect the property

- Fence and secure openings; do not enter abandoned mines or allow children or pets near pits, highwalls, or structures.
- Control erosion and dust without disturbing mine waste; obtain regulatory approval before grading or removal.

M Monitor and respond

- Watch for new depressions, cracks, cloudy or discolored water, orange staining, dead vegetation, or damaged covers.
- Test private water and relevant soil on a professional schedule and retain chain-of-custody laboratory reports.

I Prepare insurance records

- Ask specifically about mine subsidence, sinkhole, earth movement, pollution, well, and liability exclusions.
- Document secured hazards, professional assessments, reclamation, baseline condition, and required inspections.

REAL-LIFE EXAMPLE

2015 Gold King Mine Release

August 5, 2015

During work at the inactive Gold King Mine near Silverton, Colorado, a release of mine water sent millions of gallons of acidic, metal-laden water into Cement Creek and the Animas River. EPA and partner agencies conducted emergency response, monitoring, cleanup, and later Superfund work in the mining district.

Why it matters: The release illustrated how water stored behind old mine workings can mobilize contamination far downstream and why mine investigation, drainage, and emergency planning require specialized engineering.

[Open official incident record - epa.gov >](#)

Questions to resolve before making a decision

1. Search federal and state abandoned-mine inventories, geological maps, mineral records, historical aerials, and title documents.
2. Have a geotechnical professional assess suspected underground workings, subsidence, pits, fills, highwalls, and foundation implications.
3. Use qualified environmental sampling for soil, dust, water, tailings, and vapor when a credible pathway exists.
4. Determine whether reclamation obligations, access, monitoring, water-treatment, or land-use restrictions attach to the parcel.

Official sources and mapping tools

[EPA Gold King Mine questions - epa.gov >](#)[BLM Abandoned Mine Lands - blm.gov >](#)[USGS mine drainage science - usgs.gov >](#)